

UE23AM343BA2: Large Language Models and Their Applications

Jackfruit Problem – Themes.

Please don't use the below problems directly. Intent is to make write code using LLM (any LLM is OK) to solve a problem of reasonable size. Intent is to attain hands on skill in using LLM.

You can look at research paper and try to do something more to get a publishable material.

On a specific problem can try various LLMs and and get a comparative results.

The below list is what I could extract with a query.

A paper on survey of prompt engg is also here. Sufficiently large amount of coding required.

By 26 March Title and groups should be available to me. After the class. Anything discuss in class.

Evaluation: 15-23 April. Team presentation and demonstration. You can use outside class hours also.

Beginner-Friendly Projects

1. **AI Writing Assistant:** Build a tool that improves the clarity, grammar, and tone of user-provided text. This could be as simple as a web interface using an LLM API to offer suggestions.
2. **Text Summarization Tool:** Design an application that condenses long articles, documents, or meeting transcripts into concise summaries, using a framework like [LangChain](#) to process documents.
3. **Simple Q&A System Over Private Documents (RAG):** Create a system that allows users to upload a file (e.g., PDF, text file) and ask questions about its content. This uses Retrieval-Augmented Generation (RAG) to ground the model's answers in your specific data, reducing hallucinations.
4. **Email Reply Generator:** Develop a tool that takes an original email and a desired intent (e.g., "accept meeting," "decline politely") as input and generates a professional, context-aware draft response.
5. **Multilingual Translation Assistant:** Leverage the multilingual capabilities of LLMs to build a translation tool that not only translates text but can also explain the nuances and cultural context behind certain phrases.
6. **AI Story Co-Writing Assistant:** Create an interactive platform where a user provides a starting prompt, and the LLM helps generate different narrative branches, characters, or plot twists to explore creative writing possibilities.

Intermediate Projects

7. **AI Coding Assistant (Pair Programmer):** Build an assistant that can explain code snippets, generate documentation for existing codebases, or suggest improvements for a given piece of code.

8. **Sentiment Analysis and Opinion Mining Toolkit:** Develop a tool that analyzes text data (e.g., customer reviews, social media posts) and classifies the sentiment as positive, negative, or neutral, providing insights into public opinion.
9. **Job Description Analyzer/Resume Analyzer:** Create a system that extracts key requirements, skills, and responsibilities from job postings or provides suggestions to enhance a resume to match a specific job description.
10. **Custom Web Scraper for Information Extraction:** Build a smart scraper that uses LLMs to extract specific, structured information (e.g., product prices, job details, event dates) from unstructured web content, based on natural language instructions.
11. **AI Tutor for Specific Subjects:** Develop an AI tutor that explains topics like history, math, or economics using age-appropriate language and examples, tailored to a user's learning style.

Advanced Projects

12. **Text-to-SQL Query Generator:** Build an application that translates plain English business questions into functional SQL queries. This allows non-technical users to query databases using natural language.
13. **AI-Powered Financial Report Summarizer:** Create a tool that converts complex financial statements and earnings reports into plain-language summaries without adding misinterpretations or financial advice.
14. **Multi-Agent Task Automation System:** Design a system using a framework like CrewAI or AutoGen where different AI agents with specific roles (e.g., "Researcher," "Writer," "Editor") collaborate to achieve a complex goal, such as generating a complete, well-researched blog post.
15. **Build an LLM Evaluation Harness:** Develop a framework to rigorously evaluate the performance, accuracy, and consistency of different LLM outputs using metrics, confidence intervals, and automated testing.